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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

International General Certificate of Secondary Education

MARK SCHEME for the May/June 2008 question paper

0625 PHYSICS

0625/06 Paper 6 (Alternative to Practical), maximum raw mark 40

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1 **(a)** **(i)** cm, cm, g

(ii) 49.66 (or 49.7), 49.50 (or 49.5), 50.05 (or 50.0)
consistent significant figures (3 or 4) [1]

(b) clear explanation/diagram [1]

(c) correct method [1]
value 49.7 (ignore a fourth significant figure)
and allow ecf from **(ii)** [1]

(d) d = 1.8 (cm), t = 1.2 (cm) [1]
V = 3.05 (cm^3) (ecf) [1]
ρ = 16.3 unit g/cm^3, 2/3 significant figures (ecf) [1]

[Total: 9]

2 Table:

(a) Units V, A, Ω (symbol/word) [1]
R values 1.11, 2.19, 5.05, 9.55 [1]
Consistent 2 or consistent 3 sig fig for R [1]

(b) **(i)** Yes (if within 10%) No (if not) [M1]
Circuit 1 and circuit 2 compared [A1]

(ii) limit current (so temperature not increased)
OR switch off between readings
OR check for zero error
OR Repeats
OR Parallax error explained
OR Tapping meter [1]

[Total: 6]

3 Graph:
Temperature axis labelled θ/°C [1]
Suitable scales (plots occupy at least ½ grid) [1]
Plots correct to nearest ½ square (–1 each error) [2]
Lines well judged curves [1]
Lines thin [1]

(b) Statement:
larger surface area increases rate of cooling [1]
Justification:
Correct reference to gradients of lines or readings [1]

[Total: 8]

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4 Trace:

(a) all lines present, thin, neat and in correct area
normal at 90° (by eye)
and EF at 30° to normal (by eye) [1]
line KJ to at least beyond P_4 [1]

(b) (i) *a* = 12–13 (mm) no ecf [1]

(ii) *b* = 40 (mm) no ecf [1]
a and *b* both with appropriate unit [1]

(c) (i) & (ii) *c* recorded and *d* = 44 (mm) [1]

(iii) correct calculation of *n*, value 1.43 (ecf) [1]
2/3 significant figures with no unit [1]

[Total: 9]

5 (a) (i) triangle method used
(whether or not shown on graph) [1]
Triangle using more than half line
and position indicated on graph [1]
Expect *G* = 4.00–4.35 (but allow correct working
from points read from beyond 1.0 on x axis) [1]
Expect *g* = 9.07–9.87 (ecf from G) [1]

(ii) greater accuracy/average value [1]

(b) (i) amplitude [1]
length [1]
(other possible correct responses shape/size of bob
and number of swings)

(ii) does not affect time [1]

[Total: 8]